

ECO102 - Introduction to Macroeconomics

Department of Economics and Social Sciences (ESS), BRAC University

Monitoring Jobs and Inflation

Lecture 2

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 22

Outline

- Explain why unemployment is a problem, define the unemployment rate, the employment-to-population ratio, and the labor force participation rate, and describe the trends and cycles in these labor market indicators
- Explain why unemployment is an imperfect measure of underutilized labor, why it is present even at full employment, and how unemployment and real GDP fluctuate together over a business cycle
- Explain why inflation is a problem, how we measure the price level and the inflation rate, and why the CPI measure of inflation might be biased

Employment and Unemployment

Unemployment

“Definition”

A situation where individuals who are able and willing to work are unable to find suitable jobs

[Source: Google]

Unemployment results in:

- Lost incomes and production
- Lost human capital

Unemployment - Why is it a problem?

■ Lost incomes and production

- The loss of a job brings a loss of income and unemployment benefits create a safety net, but they don't fully replace lost earnings
- Lost production means lower consumption and a lower investment in capital, which lowers the living standard in both the present and the future

■ Lost human capital

- Prolonged unemployment permanently damages a person's job prospects by destroying human capital

Current Population Survey

FIGURE 22.1 Population Labor Force Categories



Population Labor Force Categories

- The **working-age population** is the total number of people aged 16 years and over who are not in jail, hospital, or some other form of institutional care
- **Labor force** is the sum of the employed and the unemployed
- To be counted as **employed** in the Current Population Survey, a person must have either a full-time job or a part-time job

Population Labor Force Categories

- To be counted as **unemployed**, a person must be available for work and must be in one of three categories:
 1. Without work but has made specific efforts to find a job within the previous four weeks
 2. Waiting to be called back to a job from which he or she has been laid off
 3. Waiting to start a new job within 30 days

- People in the working-age population who are neither employed nor unemployed are classified as not in the labor force

Population Labor Force Categories

- *In June 2010, the population of the United States was 309.6 million with the working-age population being 237.7 million. Of this number, 84 million people were in school full time or had retired from work. The remaining _____ million people made up the U.S. labor force. Of these, _____ million were employed and 14.6 million were unemployed.*

Population Labor Force Categories

- *In June 2010, the population of the United States was 309.6 million with the working-age population being 237.7 million. Of this number, 84 million people were in school full time or had retired from work. The remaining 153.7 million people made up the U.S. labor force. Of these, 139.1 million were employed and 14.6 million were unemployed.*

Three Labor Market Indicators

■ The unemployment rate

The unemployment rate is the percentage of the people in the labor force who are unemployed

$$\text{Unemployment rate} = \frac{\text{Number of people unemployed}}{\text{Labor force}} \times 100$$

Where,

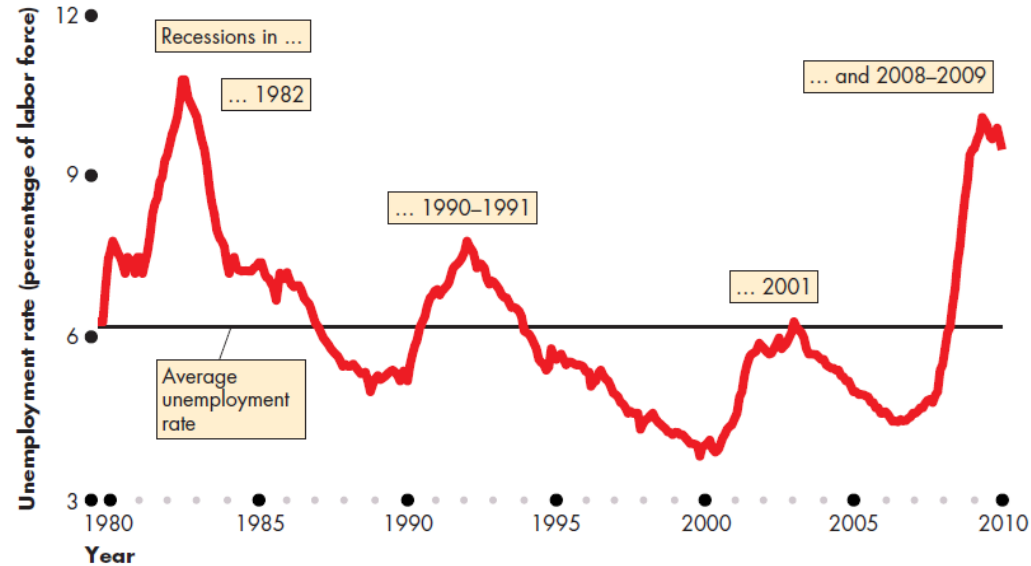
$$\text{Labor force} = \text{Number of people employed} + \text{Number of people unemployed.}$$

Calculating Unemployment Rate

- *In June 2010, the population of the United States was 309.6 million with the working-age population being 237.7 million. Of this number, 84 million people were in school full time or had retired from work. The remaining 153.7 million people made up the U.S. labor force. Of these, 139.1 million were employed and 14.6 million were unemployed.*
- *Unemployment rate = $(14.6 \div 153.7) \times 100 = 9.5\%$*

Unemployment Rate

FIGURE 22.2 The Unemployment Rate: 1980–2010



Source of data: Bureau of Labor Statistics.

Three Labor Market Indicators

■ The employment-to-population ratio

The number of people of working age who have jobs is an indicator of both the availability of jobs and the degree of match between people's skills and jobs

- The employment-to-population ratio is the percentage of people of working age who have jobs

$$\text{Employment-to-population ratio} = \frac{\text{Number of people employed}}{\text{Working-age population}} \times 100$$

Calculating Employment-to-Population Ratio

- *In June 2010, the population of the United States was 309.6 million with the working-age population being 237.7 million. Of this number, 84 million people were in school full time or had retired from work. The remaining 153.7 million people made up the U.S. labor force. Of these, 139.1 million were employed and 14.6 million were unemployed.*
- *Employment-to-population ratio = $(139.1 \div 237.7) \times 100 = 58.5\%$*

Three Labor Market Indicators

■ The labor force participation rate

The number of people in the labor force is an indicator of the willingness of people of working age to take jobs

- The labor force participation rate is the percentage of the working-age population who are members of the labor force

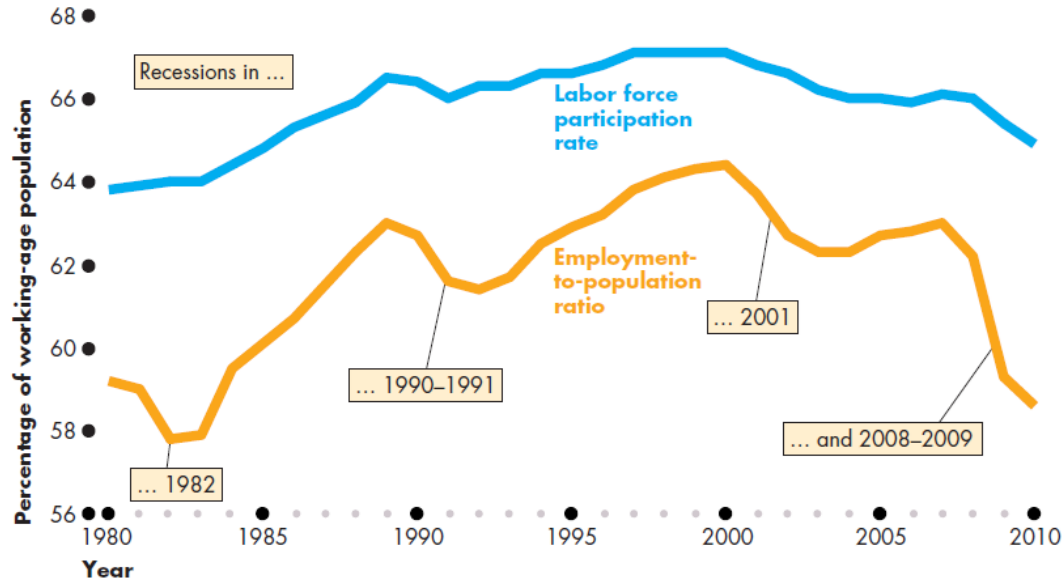
$$\text{Labor force participation rate} = \frac{\text{Labor force}}{\text{Working-age population}} \times 100.$$

Calculating Labor Force Participation Rate

- *In June 2010, the population of the United States was 309.6 million with the working-age population being 237.7 million. Of this number, 84 million people were in school full time or had retired from work. The remaining 153.7 million people made up the U.S. labor force. Of these, 139.1 million were employed and 14.6 million were unemployed.*
- *Labor force participation rate = $(153.7 \div 237.7) \times 100 = 64.7\%$*

Employment-to-Population Ratio & Labor Force Participation

FIGURE 22.3 Labor Force Participation and Employment: 1980–2010



Source of data: Bureau of Labor Statistics.

Other Definitions of Unemployment

■ Marginally attached workers

- Person who currently is neither working nor looking for work but has indicated that he or she wants and is available for a job and has looked for work sometime in the recent past
- A **discouraged worker** is a marginally attached worker who has stopped looking for a job because of repeated failure
- The official unemployment measure excludes marginally attached workers

■ Part-time workers who want full-time jobs

- Some part-time workers would like full-time jobs and can't find them
- In the official statistics, these workers are called economic part-time workers and they are partly unemployed

Most Costly Unemployment

- The costs of being unemployed for a few weeks are low compared to the costs borne by people who remain unemployed for many weeks
- Also, costs are lower for people who are unemployed because they voluntarily quit their jobs to find better ones or because they have just entered or reentered the labor market than the costs borne by people who lose their job and are forced back into the job market
- However, the unemployment rate doesn't distinguish among these different categories of unemployment

Frictional Unemployment

What is **Frictional Unemployment**?

- The unemployment that arises from the normal labor turnover that is — from people entering and leaving the labor force and from the ongoing creation and destruction of jobs—is called frictional unemployment
- Frictional unemployment is a permanent and healthy phenomenon in a dynamic, growing economy

Structural Unemployment

What is **Structural Unemployment**?

- The unemployment that arises when changes in technology or international competition change the skills needed to perform jobs or change the locations of jobs is called structural unemployment
- Structural unemployment usually lasts longer than frictional unemployment because workers must retrain and possibly relocate to find a job

Cyclical Unemployment

What is **Cyclical Unemployment**?

- The higher than normal unemployment at a business cycle trough and the lower than normal unemployment at a business cycle peak is called cyclical unemployment
- A worker who is laid off because the economy is in a recession and who gets rehired some months later when the expansion begins has experienced cyclical unemployment

“Natural” Unemployment

- **Natural unemployment** is the unemployment that arises from frictions and structural change when there is no cyclical unemployment that is — when all the unemployment is frictional and structural
- Natural unemployment as a percentage of the labor force is called the natural unemployment rate

Factors influencing Natural Unemployment Rate

■ The age distribution of the population

- An economy with a young population has a large number of new job seekers every year and has a high level of frictional unemployment
- An economy with an aging population has fewer new job seekers and a low level of frictional unemployment

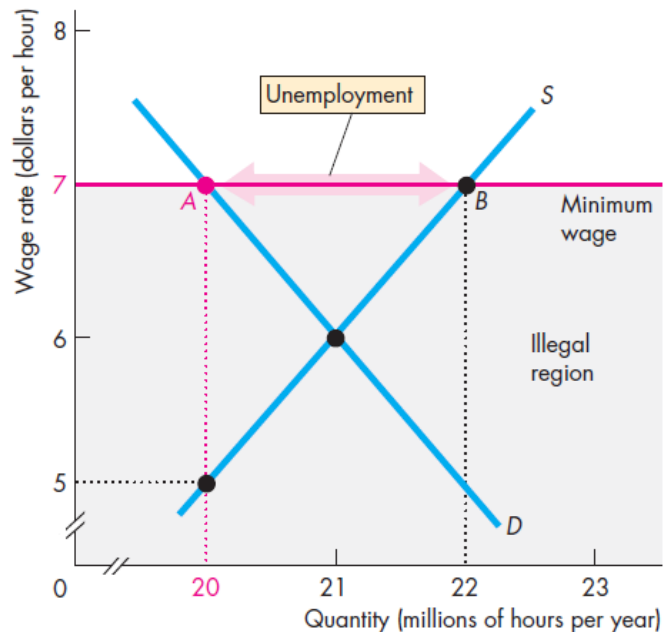
Factors influencing Natural Unemployment Rate

■ The scale of structural change

- The amount of structural unemployment fluctuates with the pace and volume of technological change and the change driven by fierce international competition, especially from fast-changing Asian economies
- When there is a technological upheaval, the old ways are swept aside and millions of jobs are lost and the skill to perform them loses value

Factors influencing Natural Unemployment Rate

■ The real wage rate



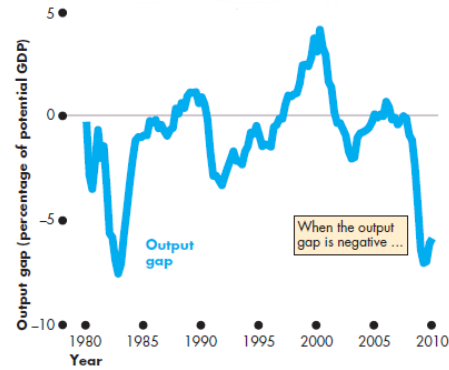
- When minimum wage rate is set above the equilibrium wage, the quantity of labor supplied exceeds the quantity of labor demanded — there is a surplus of labor
- Therefore, there is a surplus of labor
- The demand for labor determines the level of employment, and the surplus of labor is unemployed

Factors influencing Natural Unemployment Rate

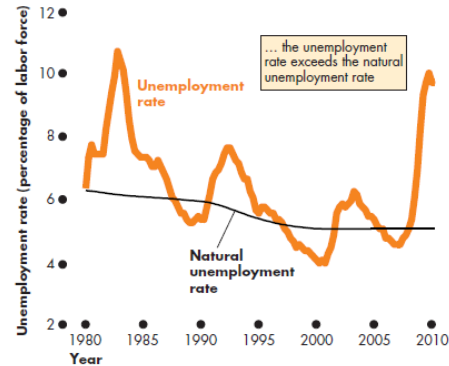
■ Unemployment benefits

- Unemployment benefits increase the natural unemployment rate by lowering the opportunity cost of job search
- Extending unemployment benefits increases the natural unemployment rate

The Output Gap and the Unemployment Rate



(a) Output gap



(b) Unemployment rate

Real GDP and Unemployment Over the Cycle

- The quantity of real GDP at full employment is potential GDP
- Over the business cycle, real GDP fluctuates around potential GDP
- The gap between real GDP and potential GDP is called the output gap
- As the output gap fluctuates over the business cycle, the unemployment rate fluctuates around the natural unemployment rate

Real GDP and Unemployment Over the Cycle

- When the economy is at full employment, the unemployment rate equals the natural unemployment rate and real GDP equals potential GDP so the output gap is zero
- When the unemployment rate is less than the natural unemployment rate, real GDP is greater than potential GDP and the output gap is positive
- And when the unemployment rate is greater than the natural unemployment rate, real GDP is less than potential GDP and the output gap is negative

The Price Level, Inflation, and Deflation

Inflation & Deflation

- **Inflation**

- A persistently rising price level is called inflation

- **Deflation**

- A persistently falling price level is called deflation

Why Inflation and Deflation are Problems

An unexpected inflation or deflation:

■ **Redistributes income**

- An unexpected burst of inflation raises prices but doesn't immediately raise the wages
- Workers are worse off because their wages buy less than they bargained for and employers are better off because their profits rise

What would be the effect of an unexpected period of deflation ?

Why Inflation and Deflation are Problems

An unexpected inflation or deflation:

■ **Redistributes wealth**

- With an unexpected burst of inflation – the borrower wins and the lender loses
- With an unexpected deflation – the borrower loses and the lender wins

■ **Lowers real GDP and employment**

- Unexpected inflation that raises firms' profits brings a rise in investment and a boom in production and employment – Real GDP rises above potential GDP and the unemployment rate falls below the natural rate
- But this situation is temporary. Profitable investment dries up, spending falls, real GDP falls below potential GDP and the unemployment rate rises
- An unexpected deflation has even greater consequences for real GDP and jobs

Why Inflation and Deflation are Problems

An unexpected inflation or deflation:

■ Diverts resources from production

- The diversion of talent/resources that results from unpredictable inflation – is a cost of inflation
- At its worst, inflation becomes hyperinflation – an inflation rate of 50 percent a month or higher that grinds the economy to a halt and causes a society to collapse

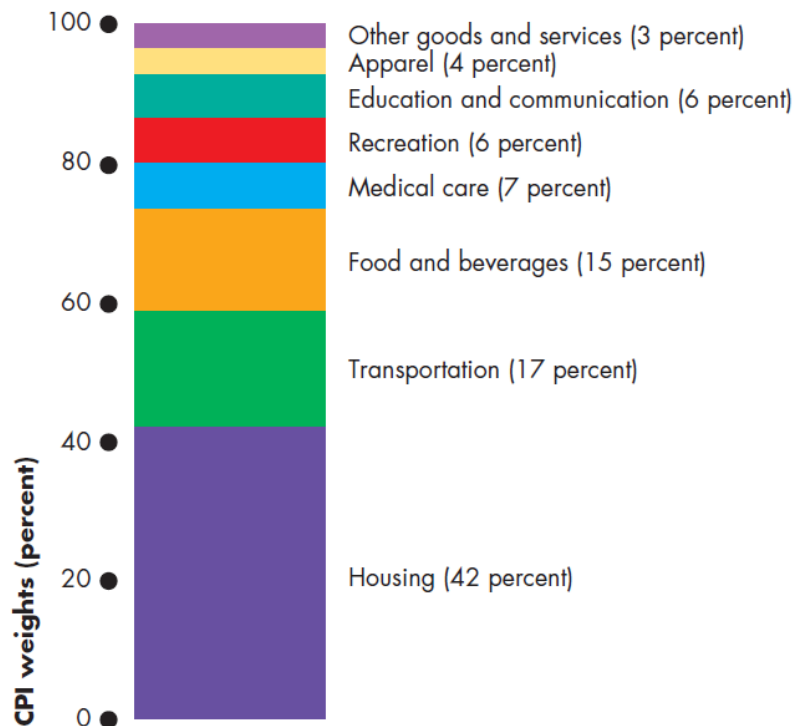
Measuring Inflation - The Consumer Price Index

Consumer Price Index (CPI) is a measure of the average of the prices paid by urban consumers for a fixed basket of consumer goods and services

Constructing the CPI involves three stages:

- Selecting the CPI basket
- Conducting the monthly price survey
- Calculating the CPI

The CPI basket



Calculating the CPI

The formula for the CPI is

$$\text{CPI} = \frac{\text{Cost of CPI basket at current prices}}{\text{Cost of CPI basket at base-period prices}} \times 100.$$

TABLE 22.1 The CPI:
A Simplified Calculation

(a) The cost of the CPI basket at base-period
prices: 2010

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$1.00	\$10
Haircuts	5	\$8.00	<u>\$40</u>
Cost of CPI basket at base-period prices			<u><u>\$50</u></u>

(b) The cost of the CPI basket at current-period
prices: 2011

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$2.00	\$20
Haircuts	5	\$10.00	<u>\$50</u>
Cost of CPI basket at current-period prices			<u><u>\$70</u></u>

Calculating the CPI

For 2010, the CPI is

$$\text{CPI} = \frac{\text{Cost of CPI basket at current prices}}{\text{Cost of CPI basket at base-period prices}} \times 100$$

$$\text{CPI in 2010} = \frac{\$50}{\$50} \times 100 = 100.$$

For 2011, the CPI is

$$\text{CPI in 2011} = \frac{\$70}{\$50} \times 100 = 140.$$

Measuring the Inflation Rate

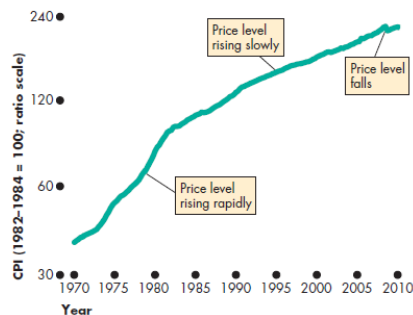
We calculate the inflation rate as the annual percentage change in the CPI

$$\text{Inflation rate} = \frac{\text{CPI this year} - \text{CPI last year}}{\text{CPI last year}} \times 100.$$

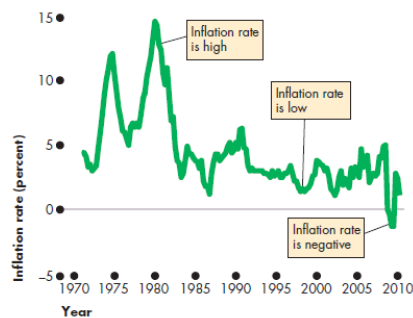
If CPI in June 2010 was 218.0, and the CPI in June 2009 was 215.7. So the inflation rate during the twelve months to June 2010 was

$$\text{Inflation rate} = \frac{(218.0 - 215.7)}{215.7} \times 100 = 1.1\%.$$

Distinguishing High Inflation from a High Price Level



(a) CPI



(b) Inflation rate

The Biased CPI

The CPI is not a perfect measure of the price level and changes in the CPI probably overstate the inflation rate

The main sources of bias in the CPI are

- New goods bias
- Quality change bias
- Commodity substitution bias
- Outlet substitution bias

The Biased CPI

The Magnitude of the Bias

- CPI overstates inflation by 1.1 percentage points a year (as per Congressional Advisory Commission)

Consequences of the Bias

- Many private agreements, such as wage contracts, that are linked to the CPI ends up giving the workers more real income than the firm intended
- Social Security checks are linked directly to the CPI and, while a bias of 1 percent a year seems small, accumulated over a decade it adds up to almost a trillion dollars of additional expenditures

Alternative Price Indexes

■ GDP deflator

It is an index of the prices of the items in consumption, investment, government expenditure, and net exports

$$\text{GDP}_{\text{deflator}} = (\text{Nominal GDP} \div \text{Real GDP}) \times 100$$